SUPPORTING INFORMATION TO ACS SENSORS

Amplification-free CRISPR/Cas12a-Based Electrochemical Biosensor with Enhanced Sensitivity for Viral Detection

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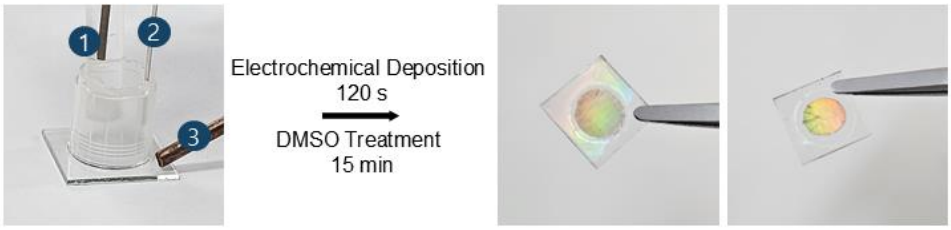


1. Reference electrode, Ag/AgCl
2. Counter electrode, Platinum
3. Working electrode, PR-patterned ITO eletrode

Figure S1. Optical images of the electrochemical deposition process using a three-electrode system and GELITION displaying holographic properties after DMSO treatment.

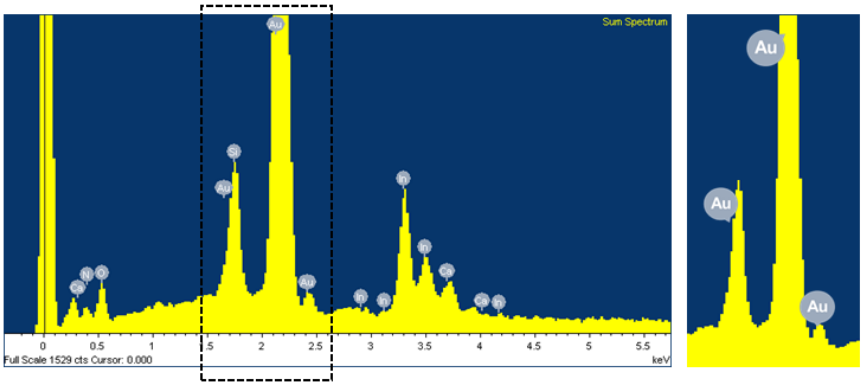


Figure S2. EDS analysis results of GELITION.

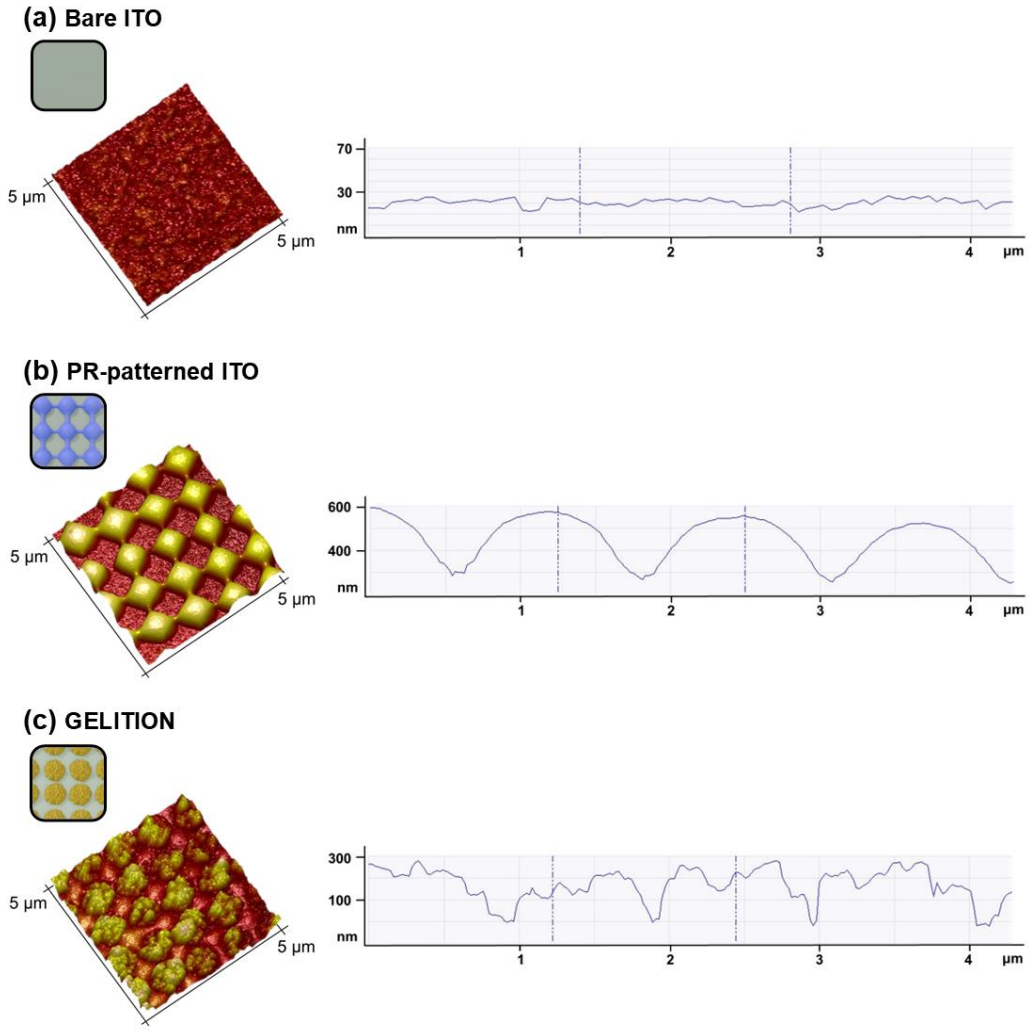


Figure S3. 3D converted AFM images with vertical investigation of (a) bare ITO electrode, (b) PR-patterned ITO electrode, and (c) GELITION.

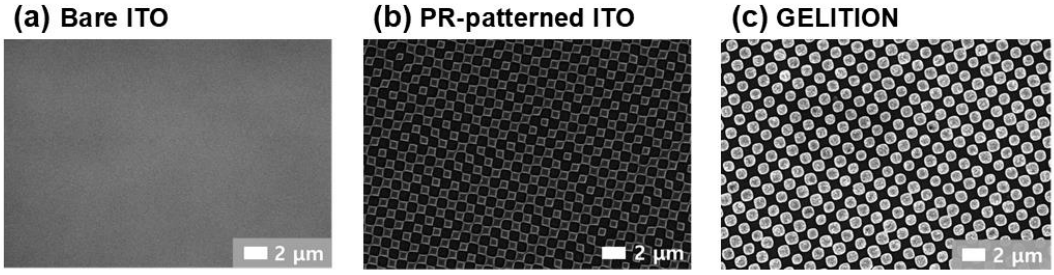


Figure S4. Zoomed-out SEM results of (a) bare ITO electrode, (b) PR-patterned ITO electrode, and (c) GELITION.

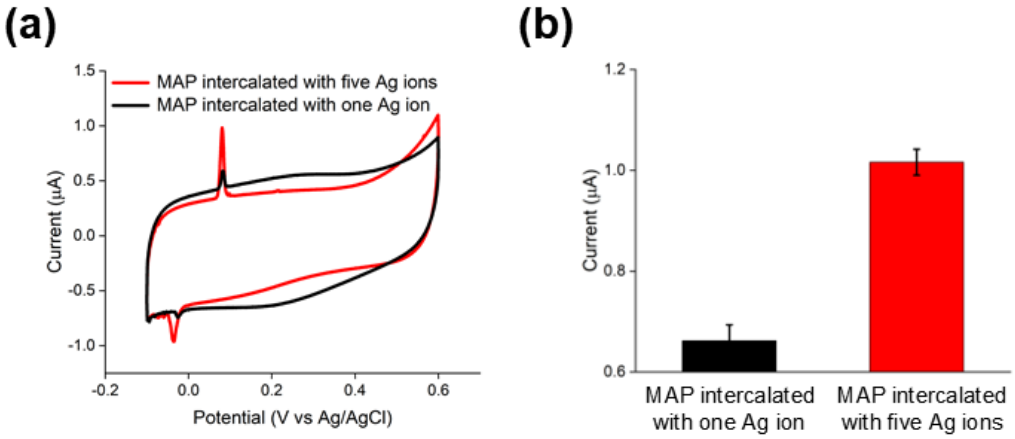


Figure S5. (a) Cyclic voltammograms and (b) oxidation peak current intensities of MAP intercalated with one Ag ion and MAP intercalated with five Ag ions. Error bars show the standard deviations of three measurements.

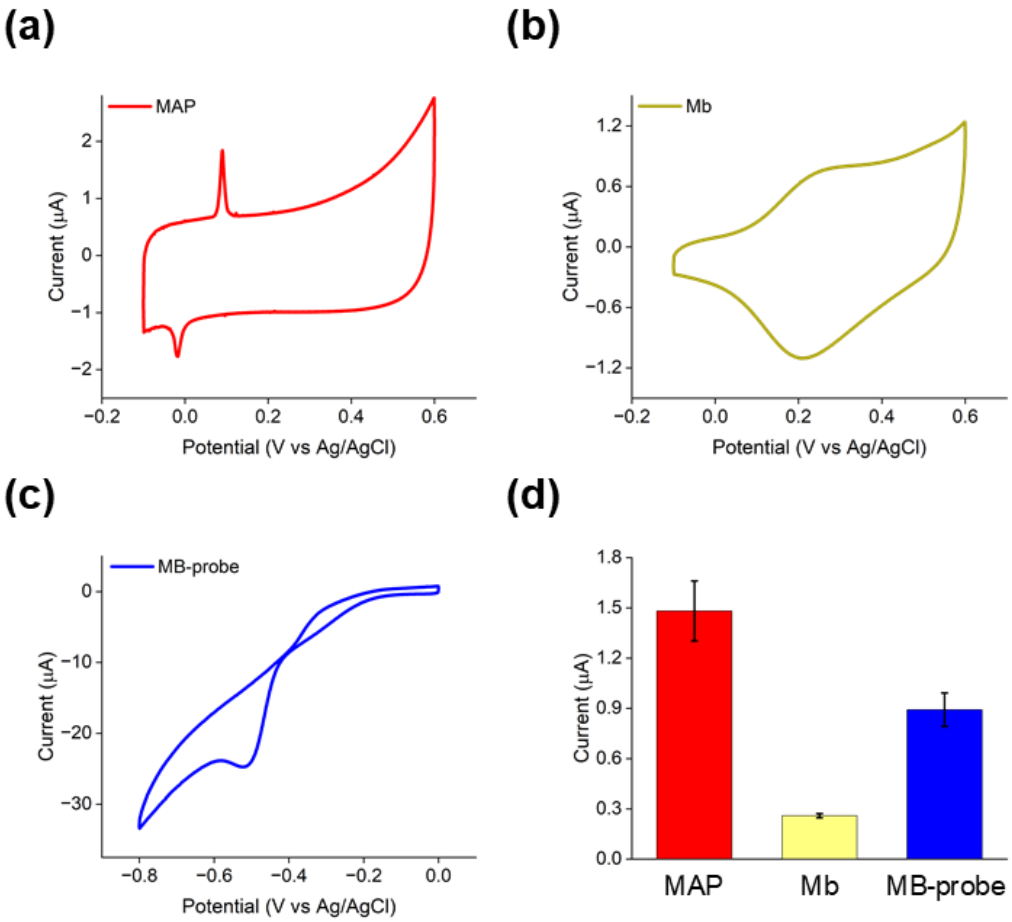


Figure S6. Cyclic voltammograms of (a) MAP, (b) Mb, and (c) MB-probe. (d) Anodic peak current (I_{pa}) intensities of the MAP, Mb, and MB-probe. Error bars show the standard deviations of three measurements.

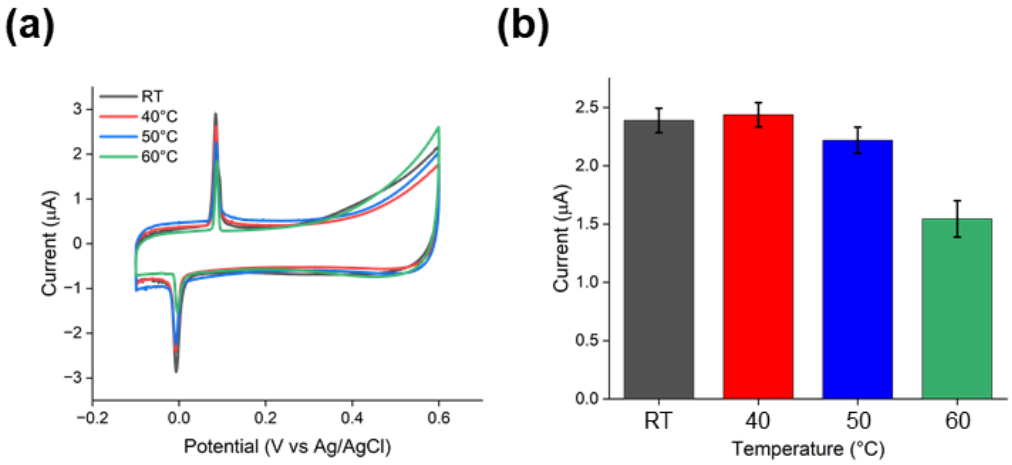


Figure S7. (a) Cyclic voltammograms and (b) oxidation peak current intensities of the MAP immobilized on the GELITION at RT, 40°C, 50°C, and 60°C. Error bars show the standard deviations of three measurements.

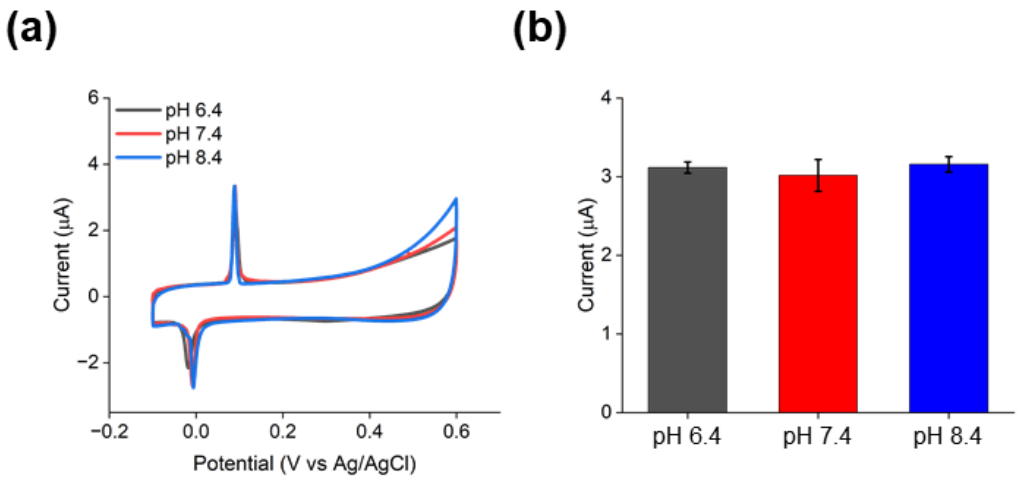


Figure S8. (a) Cyclic voltammograms and (b) oxidation peak current intensities of the MAP immobilized on the GELITION at pH 6.4, 7.4, and 8.4. Error bars show the standard deviations of three measurements.

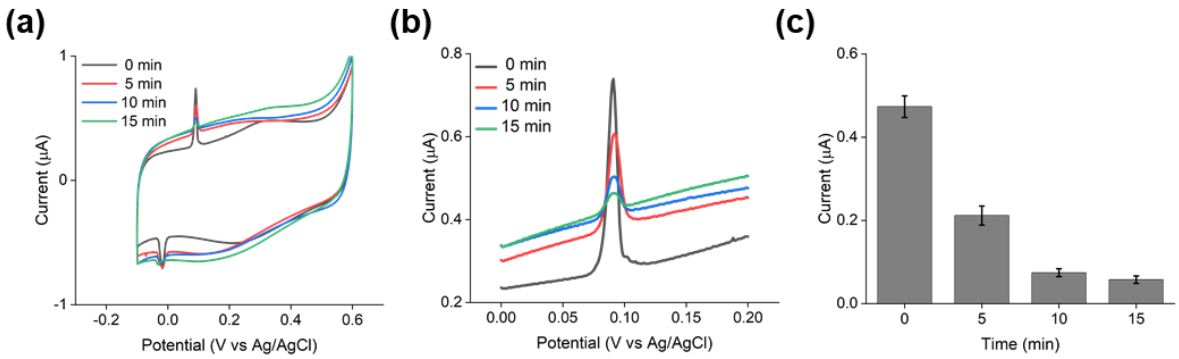


Figure S9. (a) Cyclic voltammograms, (b) zoomed-in oxidation peaks, and (c) I_{pa} intensities of MAP (at 0 min) before triplex-16 treatment and at 5, 10, and 15 min after treatment. Error bars show the standard deviations of three measurements.

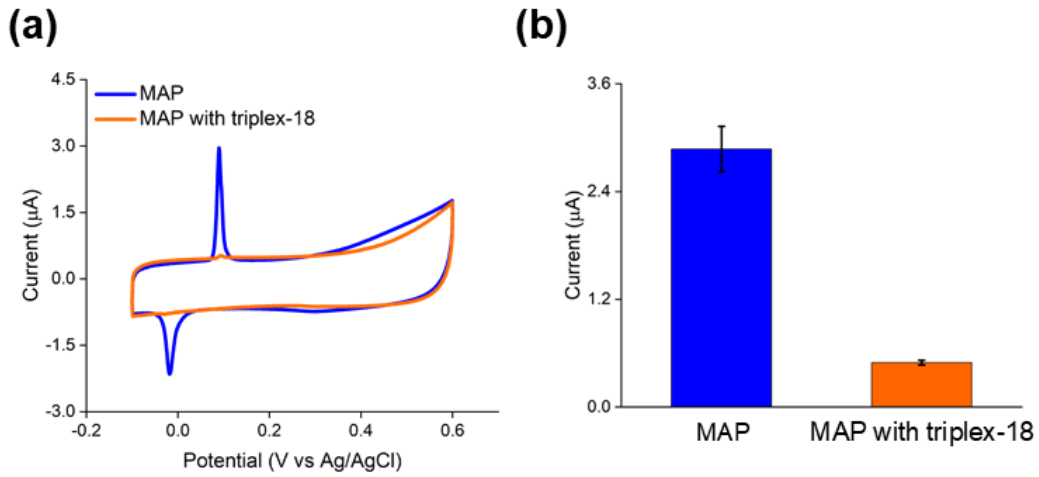


Figure S10. (a) Cyclic voltammograms and (b) oxidation peak current intensities of MAP and MAP with triplex-18. Error bars show the standard deviations of three measurements.

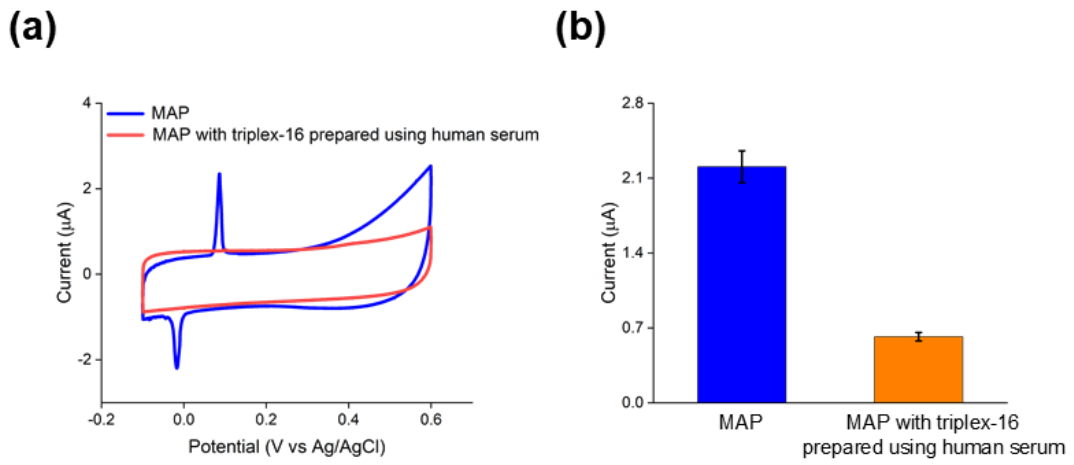
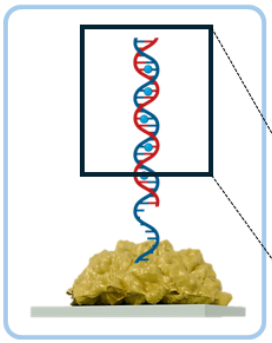
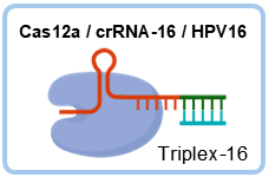


Figure S11. (a) Cyclic voltammograms and (b) oxidation peak current intensities of MAP and MAP with triplex-16 prepared using human serum. Error bars show the standard deviations of three measurements.

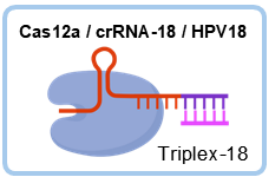


Name	Sequence (5' to 3')
MAP-L	CCC ATC GTC TTC CTC CCC CTA AAA – SH
MAP-S	GCA GCA ACA CCA TCG

C C C A T C G T C T T C C T C C C C C T A A A A - SH
G C T A C C A C A A C G A C G



Name	Sequence (5' to 3')
crRNA-16	UAA UUU CUA CUC UUG UAG AUU GAA GUA GAU AUG GCA GCA C
HPV16	GTG CTG CCA TAT CTA CTT CA



Name	Sequence (5' to 3')
crRNA-18	UAA UUU CUA CUC UUG UAG AUA CAA UAU GUG CUU CUA CAC A
HPV18	TGT GTA GAA GCA CAT ATT GT

Figure S12. The sequences of the DNA and RNA used in the development of an amplification-free CRISPR/Cas12a-based electrochemical biosensor.